MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT – 10** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions

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Q1. For any acute angle θ, the value of $\sin^2\theta + \cos^2\theta$ is always: [1]

(A) Less than 1
(B) Equal to 1
(C) Greater than 1
(D) Depends on θ

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) Equal to 1

Justification: By the fundamental Pythagorean identity in a right-angled triangle, $\sin^2\theta + \cos^2\theta = \dfrac{P^2 + B^2}{H^2} = \dfrac{H^2}{H^2} = 1$.

Q2. As the acute angle θ increases from 0° to 90°, the value of $\cos\theta$: [1]

(A) Increases from 0 to 1
(B) Decreases from 1 to 0
(C) Remains constant
(D) Increases then decreases

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) Decreases from 1 to 0

Justification: Standard values: $\cos 0^\circ = 1$, $\cos 30^\circ = \dfrac{\sqrt{3}}{2}$, $\cos 45^\circ = \dfrac{1}{\sqrt{2}}$, $\cos 60^\circ = \dfrac{1}{2}$, $\cos 90^\circ = 0$.

Q3. If $\tan\theta + \cot\theta = 2$, then the value of $\tan^{10}\theta + \cot^{10}\theta$ is: [1]

(A) 1 (B) 2 (C) 2^{10} (D) 20

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 2

Justification: $\tan\theta + \dfrac{1}{\tan\theta} = 2 \implies (\tan\theta - 1)^2 = 0 \implies \tan\theta = \cot\theta = 1$. Thus $1^{10} + 1^{10} = 2$.

Q4. If $x = 2\sin^2\theta$ and $y = 2\cos^2\theta + 1$, then the value of $x + y$ is: [1]

(A) 2 (B) 3 (C) 1 (D) 5

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 3

Justification: $x + y = 2(\sin^2\theta + \cos^2\theta) + 1 = 2(1) + 1 = 3$.

Q5. If $\triangle ABC$ is an equilateral triangle, then the value of $\sin A + \cos B$ is: [1]

(A) $\dfrac{\sqrt{3}+1}{2}$ (B) $\sqrt{3}$

(C) 1 (D) $\dfrac{\sqrt{3}}{2}$

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) $\dfrac{\sqrt{3}+1}{2}$

Justification: In equilateral $\triangle ABC$, $A = B = 60°$. Hence $\sin 60° + \cos 60° = \dfrac{\sqrt{3}}{2} + \dfrac{1}{2} = \dfrac{\sqrt{3}+1}{2}$.

Q6. **Assertion (A):** If $\sec\theta + \tan\theta = 5$, then $\sec\theta - \tan\theta = 0.2$.
Reason (R): $\sec^2\theta - \tan^2\theta = 1 \implies (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Justification: From (R), $\sec\theta - \tan\theta = \dfrac{1}{5} = 0.2$. Both statements are true and (R) correctly explains (A).

Q7. Evaluate: $\dfrac{\cos 45^\circ}{\sec 30^\circ + \csc 30^\circ}$. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Substitute Standard Angle Values [1 Mark]

$$\frac{1/\sqrt{2}}{\frac{2}{\sqrt{3}} + 2} = \frac{1/\sqrt{2}}{\frac{2(1+\sqrt{3})}{\sqrt{3}}} = \frac{\sqrt{3}}{2\sqrt{2}(\sqrt{3}+1)}$$

Step 2: Rationalize Denominator [1 Mark]

$$= \frac{\sqrt{3}(\sqrt{3}-1)\sqrt{2}}{2\sqrt{2}\cdot\sqrt{2}\cdot(3-1)} = \boxed{\frac{\mathbf{3\sqrt{2}-\sqrt{6}}}{\mathbf{8}}}$$

Q8. If $\tan\theta + \cot\theta = 2$, find the value of $\tan^2\theta + \cot^2\theta$. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Square Both Sides of the Given Equation [1 Mark]

$$(\tan\theta + \cot\theta)^2 = 2^2 \implies \tan^2\theta + \cot^2\theta + 2\tan\theta\cot\theta = 4$$

Step 2: Substitute $\tan\theta\cot\theta = 1$ [1 Mark]

$$\tan^2\theta + \cot^2\theta + 2(1) = 4 \implies \boxed{\tan^2\theta + \cot^2\theta = \mathbf{2}}$$

Q9. Prove that: $\dfrac{\cot A - \cos A}{\cot A + \cos A} = \dfrac{\csc A - 1}{\csc A + 1}$. [3]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Write $\cot A = \cos A \csc A$ [1.5 Marks]

$$\text{LHS} = \frac{\frac{\cos A}{\sin A} - \cos A}{\frac{\cos A}{\sin A} + \cos A} = \frac{\cos A\csc A - \cos A}{\cos A\csc A + \cos A}$$

Step 2: Factor Out $\cos A$ **& Cancel** $(\cos A \neq 0)$ [1.5 Marks]

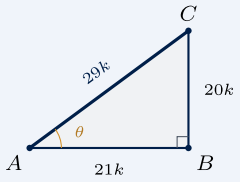
$$\text{LHS} = \frac{\cos A(\csc A - 1)}{\cos A(\csc A + 1)} = \boxed{\frac{\csc \mathbf{A} - \mathbf{1}}{\csc \mathbf{A} + \mathbf{1}} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q10. If $\tan\theta = \dfrac{20}{21}$, evaluate $\dfrac{1-\sin\theta+\cos\theta}{1+\sin\theta+\cos\theta}$. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Find $\sin\theta$ and $\cos\theta$ via Right Triangle [1.5 Marks]

Let $\triangle ABC$ ($\angle B = 90°, \angle A = \theta$) have $BC = 20k$, $AB = 21k$ ($k > 0$).



$$AC = \sqrt{(20k)^2 + (21k)^2} = \sqrt{400k^2 + 441k^2} = \sqrt{841k^2} = 29k$$

Hence $\sin\theta = \dfrac{20}{29}$ and $\cos\theta = \dfrac{21}{29}$.

Step 2: Substitute & Evaluate [1.5 Marks]

$$\frac{1-\frac{20}{29}+\frac{21}{29}}{1+\frac{20}{29}+\frac{21}{29}} = \frac{\frac{29-20+21}{29}}{\frac{29+20+21}{29}} = \frac{30}{70} = \boxed{\frac{3}{7}}$$

SECTION D: Long Answer / HOTS

MATHS MANTHAN | **(2 × 5 = 10 Marks)**

Q11. If $\csc\theta - \sin\theta = l$ and $\sec\theta - \cos\theta = m$, prove that $l^2m^2(l^2+m^2+3) = 1$. **[5]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Step 1: Simplify l and m in Terms of $\sin\theta, \cos\theta$ [1.5 Marks]

$$l = \frac{1}{\sin\theta} - \sin\theta = \frac{1-\sin^2\theta}{\sin\theta} = \frac{\cos^2\theta}{\sin\theta}, \qquad m = \frac{1}{\cos\theta} - \cos\theta = \frac{1-\cos^2\theta}{\cos\theta} = \frac{\sin^2\theta}{\cos\theta}$$

Step 2: Evaluate l^2m^2 and (l^2+m^2+3) [2 Marks]

$$l^2m^2 = \left(\frac{\cos^2\theta}{\sin\theta}\cdot\frac{\sin^2\theta}{\cos\theta}\right)^2 = (\sin\theta\cos\theta)^2 = \sin^2\theta\cos^2\theta$$

$$l^2+m^2+3 = \frac{\cos^4\theta}{\sin^2\theta} + \frac{\sin^4\theta}{\cos^2\theta} + 3 = \frac{\cos^6\theta+\sin^6\theta+3\sin^2\theta\cos^2\theta}{\sin^2\theta\cos^2\theta}$$

Step 3: Apply $a^3+b^3+3ab(a+b) = (a+b)^3$ with $a = \cos^2\theta, b = \sin^2\theta$ [1.5 Marks]

$$l^2m^2(l^2+m^2+3) = \sin^2\theta\cos^2\theta \cdot \frac{(\cos^2\theta+\sin^2\theta)^3}{\sin^2\theta\cos^2\theta} = 1^3 = \boxed{1 \quad \text{[Hence Proved]}}$$

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Q12. If $x = r\sin A\cos C$, $y = r\sin A\sin C$, and $z = r\cos A$, prove that $r^2 = x^2 + y^2 + z^2$. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Compute x^2, y^2, and z^2 [1.5 Marks]

$$x^2 = r^2\sin^2 A\cos^2 C, \qquad y^2 = r^2\sin^2 A\sin^2 C, \qquad z^2 = r^2\cos^2 A$$

Step 2: Sum $x^2 + y^2$ Using $\cos^2 C + \sin^2 C = 1$ [2 Marks]

$$x^2 + y^2 = r^2\sin^2 A(\cos^2 C + \sin^2 C) = r^2\sin^2 A(1) = r^2\sin^2 A$$

Step 3: Add z^2 Using $\sin^2 A + \cos^2 A = 1$ [1.5 Marks]

$$x^2 + y^2 + z^2 = r^2\sin^2 A + r^2\cos^2 A = r^2(\sin^2 A + \cos^2 A) = \boxed{\mathbf{r^2} \quad \text{[Hence Proved]}}$$

SECTION E: Case-Based / Applied Situation — MATHS MANTHAN | (1 × 4 = 4 Marks)

Q13. Retaining Wall — Earth Embankment Batter Slope

A civil engineering department constructs a retaining wall with a front batter face inclined to prevent earth slippage. The cross-section forms a right-angled triangle where the vertical wall height is $h = 4$ m, and the face is sloped at an angle of $\theta = 45^\circ$ to the horizontal foundation bed. **Based on the above geometric model, answer the following questions:**

(i) State the value of $\tan 45^\circ$ and find the horizontal batter spread b of the foundation. [1]

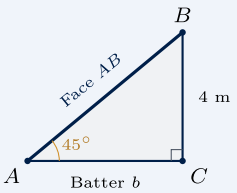
(ii) Determine the length of the diagonal inclined face of the retaining wall. [1]

(iii) Verify that $\sin^2 45^\circ + \cos^2 45^\circ = 1$ using the dimensions of this structure. [2]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Sub-part (i): $\tan 45^\circ$ and Horizontal Spread b [1 Mark]

$$\tan 45^\circ = \mathbf{1}, \qquad \tan 45^\circ = \frac{BC}{AC} \implies 1 = \frac{4}{b} \implies \boxed{b = \mathbf{4}\text{ m}}$$



Sub-part (ii): Diagonal Inclined Face AB [1 Mark]

$$AB = \sqrt{AC^2 + BC^2} = \sqrt{4^2 + 4^2} = \sqrt{32} = \boxed{\mathbf{4\sqrt{2}}\text{ m} \approx 5.66\text{ m}}$$

Sub-part (iii): Identity Verification via Structure Dimensions [2 Marks]

$$\sin 45^\circ = \frac{BC}{AB} = \frac{4}{4\sqrt{2}} = \frac{1}{\sqrt{2}}, \qquad \cos 45^\circ = \frac{AC}{AB} = \frac{4}{4\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\sin^2 45^\circ + \cos^2 45^\circ = \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} + \frac{1}{2} = \boxed{\mathbf{1} \quad \text{(Verified)}}$$

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